Report

Course	2nd Year Physiology 2020
Lesson	Cardiovascular Physiology – Practical
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Experiment 1a: A Normal ECG

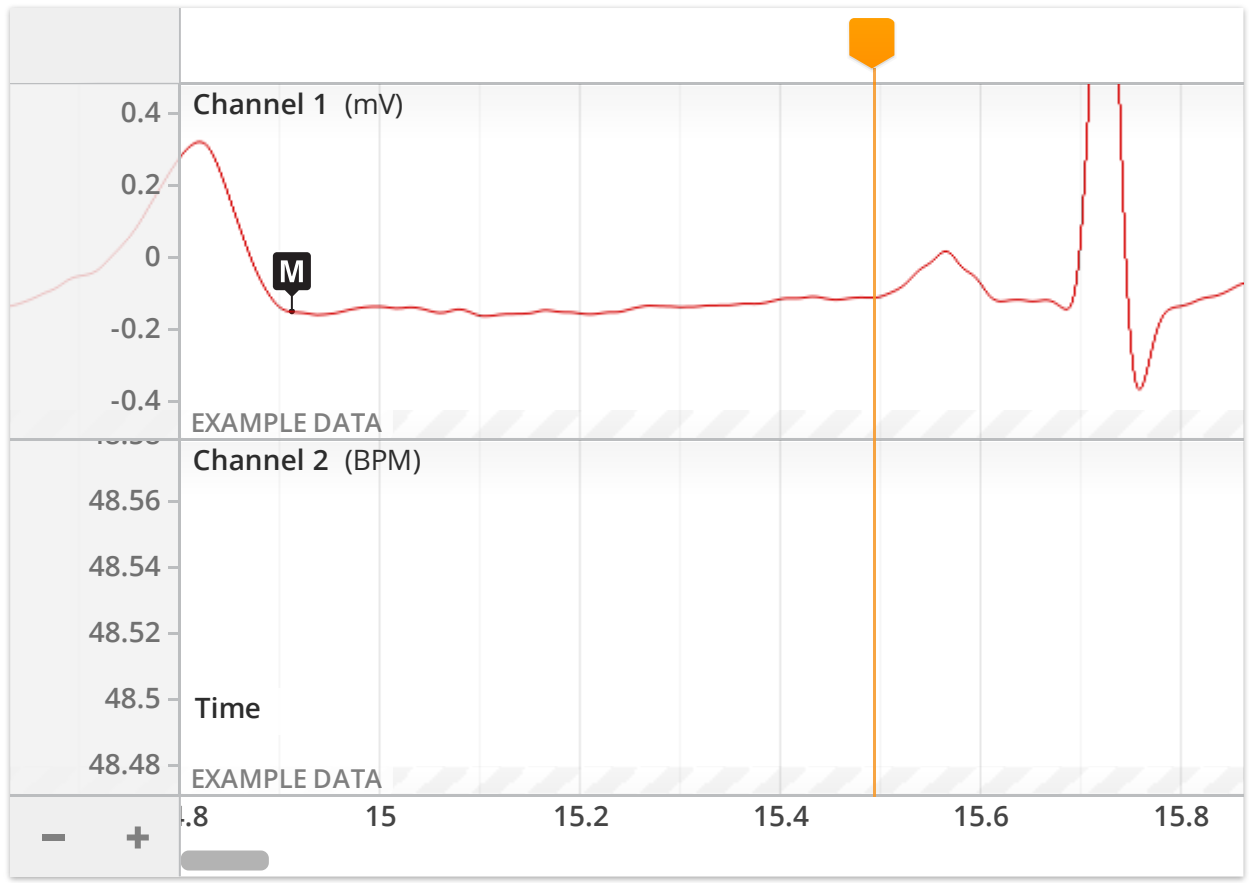


Table 1. ECG Component Durations

Component	R-R Interval	T-P Interval	P-R Interval	P-wave	QRS Complex	T-wave
Duration (msec)	1190	560				

Calculate the heart rate from the R-R interval given in table 1.

R-R interval is 1.19 seconds therefore it took 1.19 seconds for one heartbeat to start and to finish. The units for heart rate is beats per minute. One minute=60 seconds. Therefore over 60 seconds we have 50.42 beats. Meaning, 50.42 beats per minute.

Heart rate= 50.42 BPM

Experiment 1b - A Normal ECG

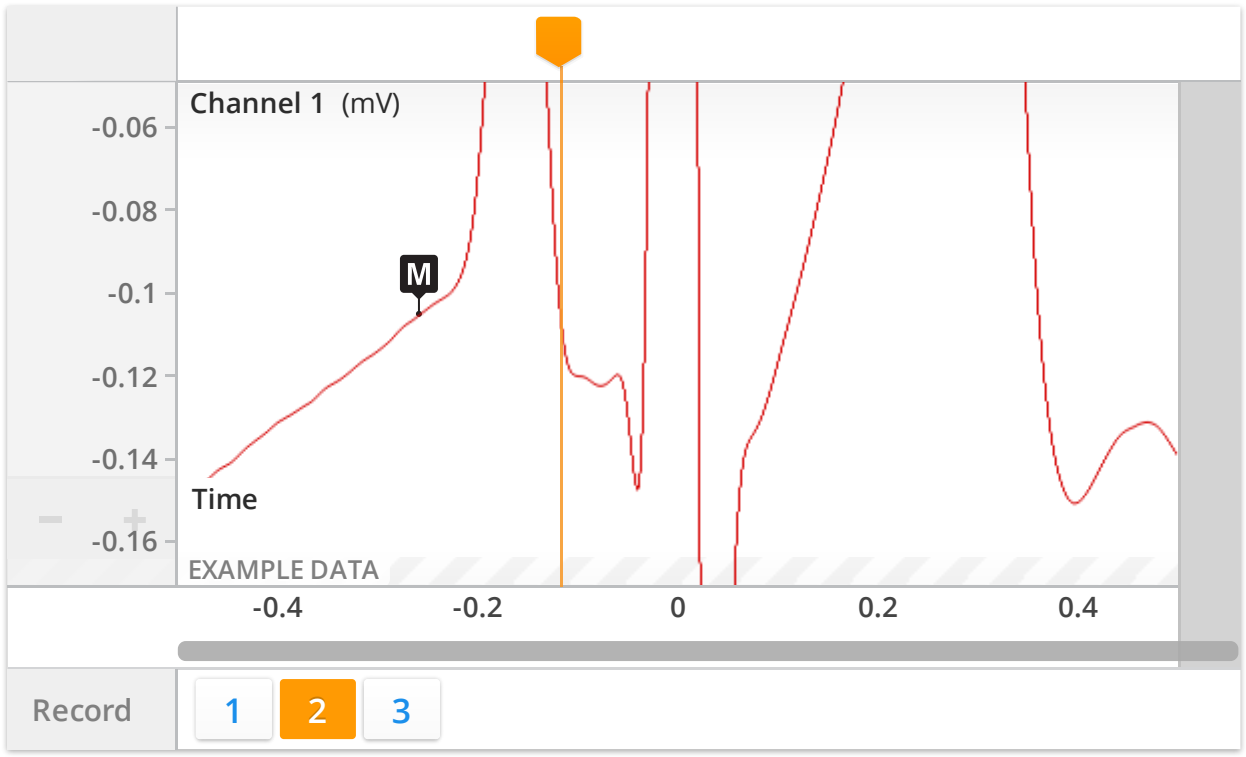


Table 1. ECG Component Durations

Component	R-R Interval	T-P Interval	P-R Interval	P-wave	QRS Complex	T-wave
Duration (msec)	1190.00	560.00	250	140	120	190

Experiment 2 - Respiratory Arrhythmia

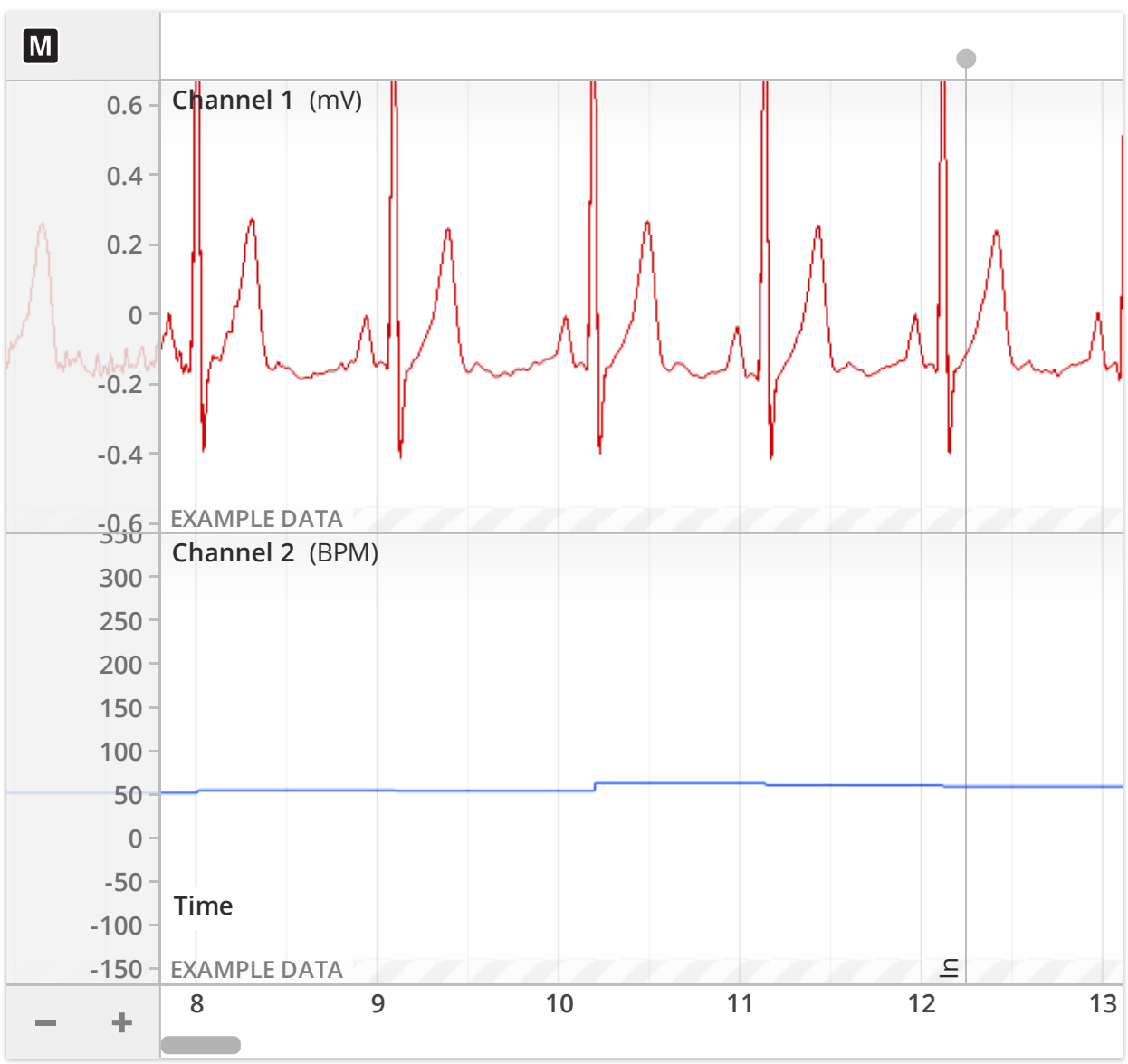


Table 2. Effect of different breathing patterns on heart rate.

Breathing	Minimum HR (BPM)	Maximum HR (BPM)	Average HR (BPM)
Normal	46.28	56.18	49.83
Exaggerated	52.27	86.16	66.49

Did heart rate increase during inspiration or expiration?

While breathing in or during inspiration, the heart rate will increase. While breathing out, or expiration the heart rate decreases.

Propose a mechanism for the relationship between heart rate and ventilation.

When we breathe in, we are taking in more oxygen through our lungs. Therefore, the heart rate will increase to deliver the oxygen to the respiring muscles in a more efficient manner.

Also during inspiration, the thorax contracts to allow the lungs to expand, as a result, these contracting muscles need more oxygen delivered to them. Hence, the heart rate increases to allow the thorax to contract when we breathe in.

However, when we breathe out, the thorax muscles contract as the lungs no longer need to expand. Because of the reduced need for oxygen being delivered to these muscles when they are relaxing, the heart rate slightly drops.

Experiment 3 - Electrical Interference of Skeletal Muscle Contraction on ECG

1 pt

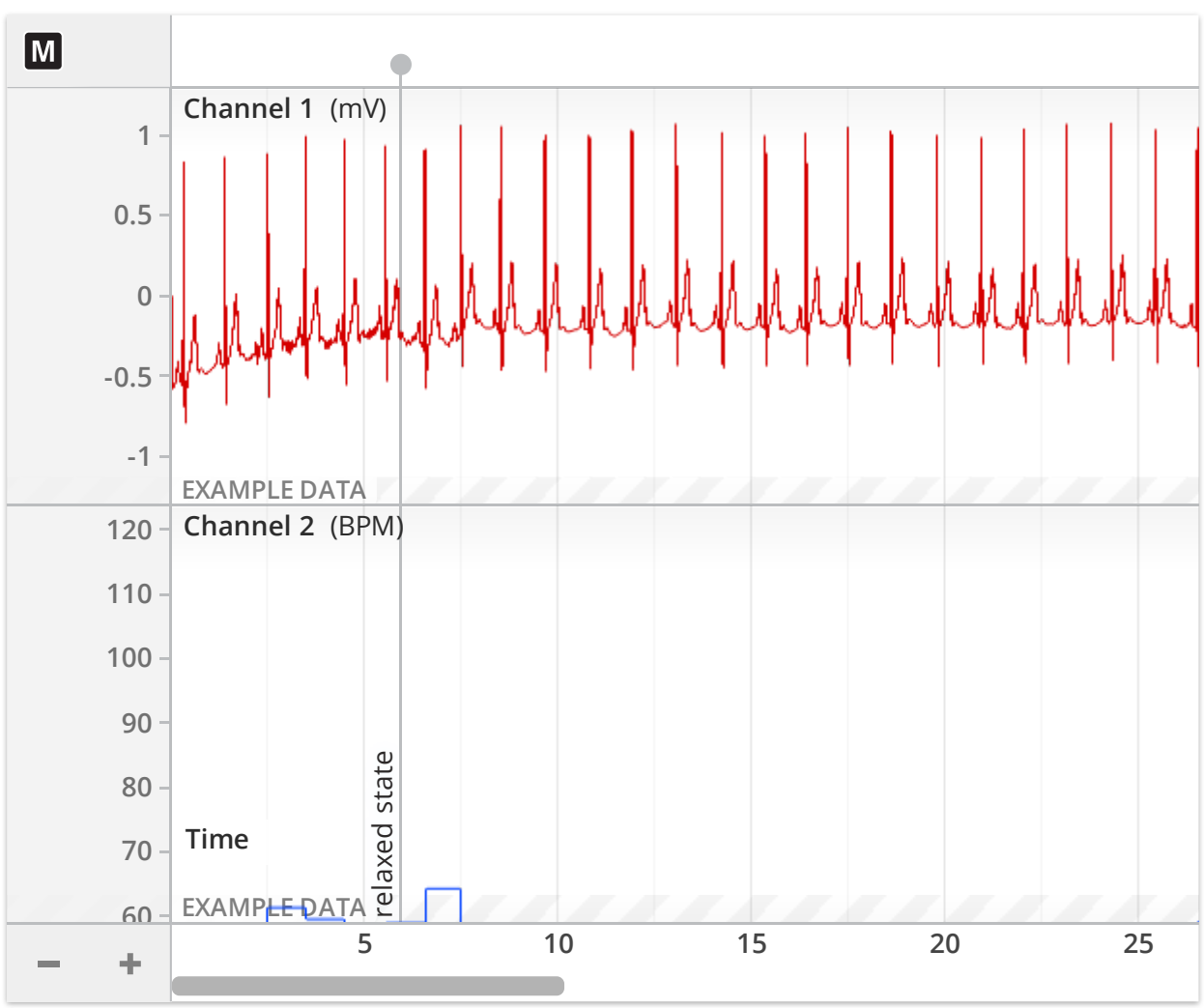


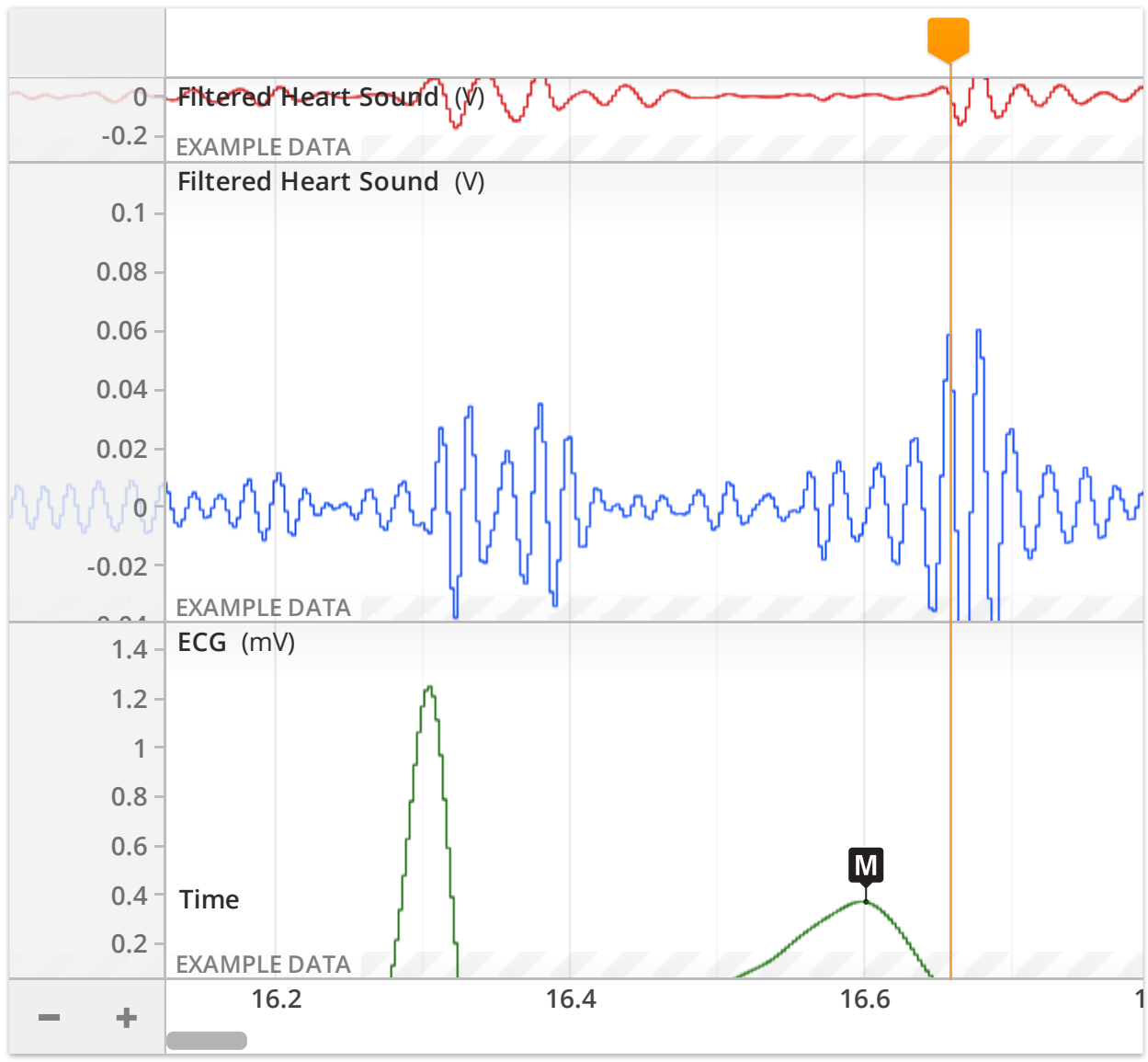
Table 3. Mean heart rate before and after skeletal muscle contraction

Relaxed State (BPM)	Muscle Contraction (BPM)
63.36	74.91

What changes did you observe in the ECG recording during skeletal muscle activity and why?

Upon muscle contraction, the heart rate increases. It has been suggested that this may be due to the muscle mechanoreceptors which selectively decrease parasympathetic output, and therefore increase the heart rate. This may be in a response for the need to carry oxygen more quickly to the respiring skeletal muscles.

Experiment 4 - Heart Sounds



Experiment 4 - Heart Sounds

Table 4. ECG and Phonocardiography (PCG)

R to first sound (ms)	T to second sound (ms)
7.5	57.5

The first heart sound:

When the ventricles begin to start contracting, or ventricular depolarisation, during isovolumic ventricular contraction, the blood flow causes the mitral and tricuspid valves to close. The first heart sounds happen after the mitral and the tricuspid valve close. The shutting of the valves result in a turbulent flow of blood within the ventricular compartment of the heart. The interaction between the turbulent blood flow that the walls of the chamber cause vibrations. These vibrations are heart as the first heart sound.

The second heart sound:

As pressure within the ventricular compartments build up as ventricular depolarisation continues to increase during isovolumic contraction, the semilunar valves open and blood from the right ventricles enter the pulmonary artery and blood from the left ventricle enters the aorta. This brings us to the end of isovolumic ventricular contraction and the start of ventricular ejection. At this point depolarisation continues after the end systolic volume in the ventricles remains. This is followed by ventricular repolarisation. Since electrical activity shortly precedes mechanical activity, isovolumic ventricular relaxation happens shortly after. It is at this point at which the blood flows back into the cusps of the semilunar valves and closes them shut. This causes a turbulent flow within the aorta and the pulmonary artery. The interaction between such a blood flow and the walls of the vessels cause vibrations which translates to the second heart sound.

Experiment 5 - Measuring HR and BP

Table 1: Blood pressure and heart rate measurements

Subject #	Systolic Pressure	Diastolic Pressure	Heart Rate	MAP
	( )	( )	( )	( )
Subject 1	96	62	58	73.3
Subject 1	99	61	63	73.7
Subject 2	120	68	52	85.3
Subject 2	118	75	53	89.3
Subject 3				
Subject 3				

1. Why is it the convention to measure blood pressure in the upper arm, hanging by your side? Why not the foot, or with your arm up in the air? Think about what you want to measure and how that might be different if measured elsewhere in the body.

The upper arm is close to the heart therefore little pressure is lost to gravity at this point. Also the blood at this point would have picked up very little friction as it wouldn't have had to flow very far to get to this point. Therefore, the overall pressure loss at this point is not significant.

*In the foot, the blood would have picked up alot of pressure and with the arm in the air gravity would decelerate the flow of blood along the arm. Both these situations would cause a significant drop in pressure.

2. From your observed measurements, what was the pulse pressure? Show your calculations.

1/1: 96-62= 34

1/2: 99-61= 38

2/1: 120-68= 52

2/2: 118-75= 43

3. What does MAP represent and what are the two factors that contribute to MAP in the body? Summarise the physiology of how these factors can be changed/altered.

2 main factors that contribute to the mean arterial pressure: Cardiac output and the total peripheral resistance within the system

Increase the cardiac output, that should increase mean arterial pressure and if we decrease the cardiac output that should increase our mean arterial pressure

More resistance in the arterial side of the system is going to increase mean arterial pressure; vice versa= little inhibition

Different factors that contribute to cardiac output: Heart rate and stroke volume

-To alter cardiac output we will generally increase heart rate and that primarily links to ejecting more volume from the heart

TPR:

- Alter resistance through metabolic demand

-How viscous our blood is

-Lengths of our vascular system

*Bigger the diameter of the radius= the less the radius

Dynamic and Static Exercise

List each exercise that you have chosen in the table below.

Dynamic Exercise	Static Exercise
30 second skipping	30 second plank

Dynamic Exercise

Baseline measurements for dynamic exercise.

Dynamic exercise	Heart rate
Baseline 1	65
Baseline 2	69
Baseline 3	66
Baseline 4	69
Baseline 5	72

Exercise heart rate measurements for dynamic exercise.

Dynamic exercise	Heart rate
Effort 1	92
Effort 2	95
Effort 3	98
Effort 4	97
Effort 5	101

Recovery heart rates for dynamic exercise

Dynamic Exercise	Heart rate (1 min post exercise)	Heart rate (3 min post exercise)
Recovery 1	85	70
Recovery 2	84	72
Recovery 3	87	79
Recovery 4	91	78
Recovery 5	93	77

Static Exercise

Baseline measurements for static exercise.

Static exercise	Heart rate
Baseline 1	68
Baseline 2	65
Baseline 3	66
Baseline 4	69
Baseline 5	66

Exercise measurements for static exercise.

Static exercise	Heart rate
Effort 1	98
Effort 2	103
Effort 3	101
Effort 4	100
Effort 5	104

Recovery heart rates for static exercise.

Static Exercise	Heart rate (1 min post exercise)	Heart rate (3 min post exercise)
Recovery 1	70	66
Recovery 2	72	65
Recovery 3	71	68
Recovery 4	73	69
Recovery 5	75	68

Evaluation

In a few words indicate what you most enjoyed in this Practical:

In a few words indicate what could be improved in this Practical: